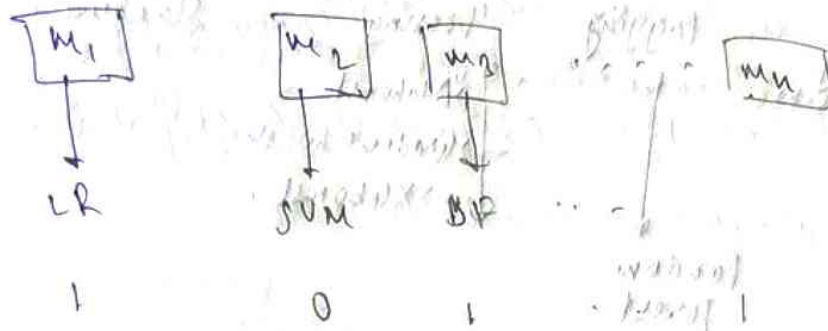


Ensemble learning:

Predictions is done using a combination of n models (base) and ~~for~~ prediction using majority voting.



Assume, n models

- All have diff ~~models~~ algos (LR, Df, SVM, ...)
- All have same algo but diff data ($D_1, D_2, \dots$)

Classification

for 5 ~~algo~~ models

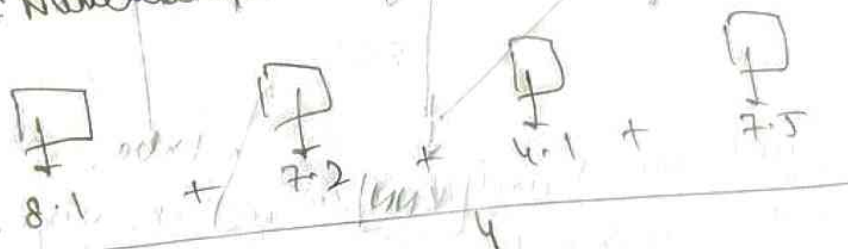
gpa / iq / Percent



$y \in \{0, 1\}$ $\rightarrow$ $\begin{cases} 3 \rightarrow 1 \\ 2 \rightarrow 0 \end{cases}$ $\rightarrow$ by majority, $1 \rightarrow 1$

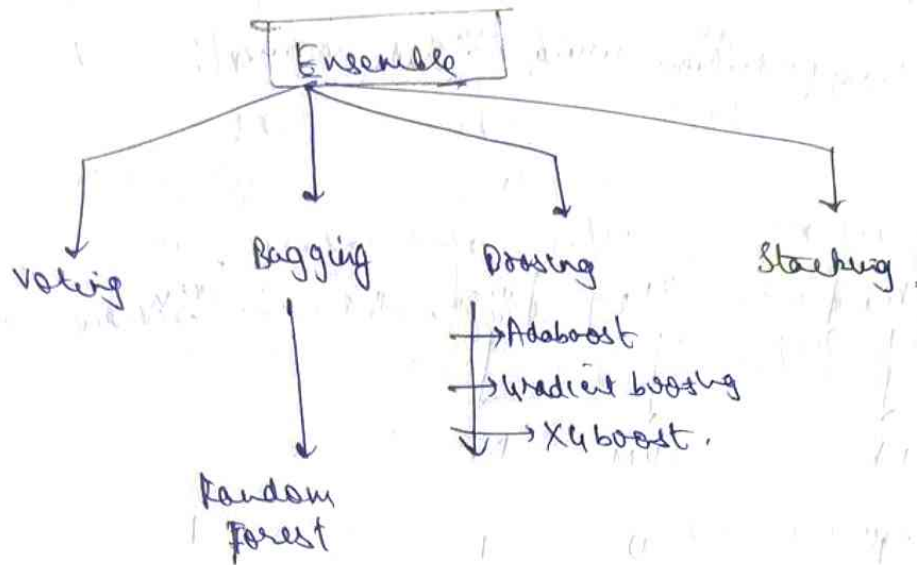
Regression

: Numerical O/P.

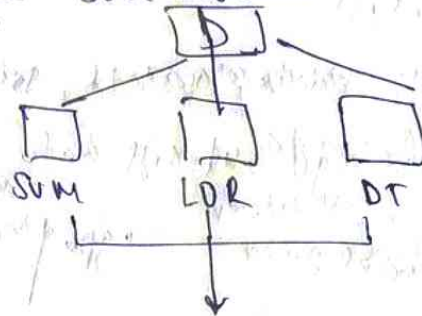


(Calculate mean of all base models (prediction))

Type of ensemble:



Voting Ensemble: Base models are diff algos and same

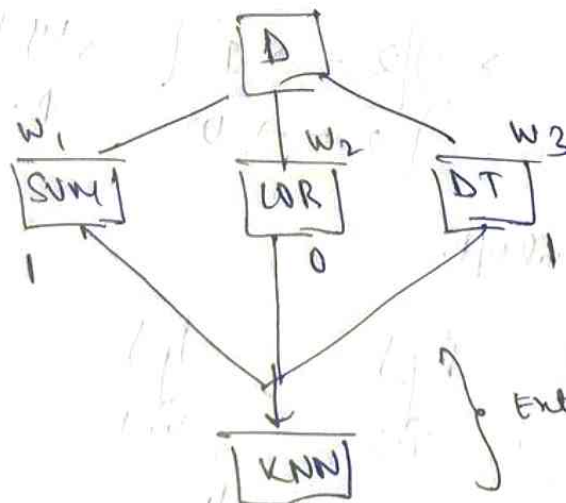


$\{Xq\}$

↓
Data point

$\{Xq\}$

Stacking: Similar to voting, contains an extra model



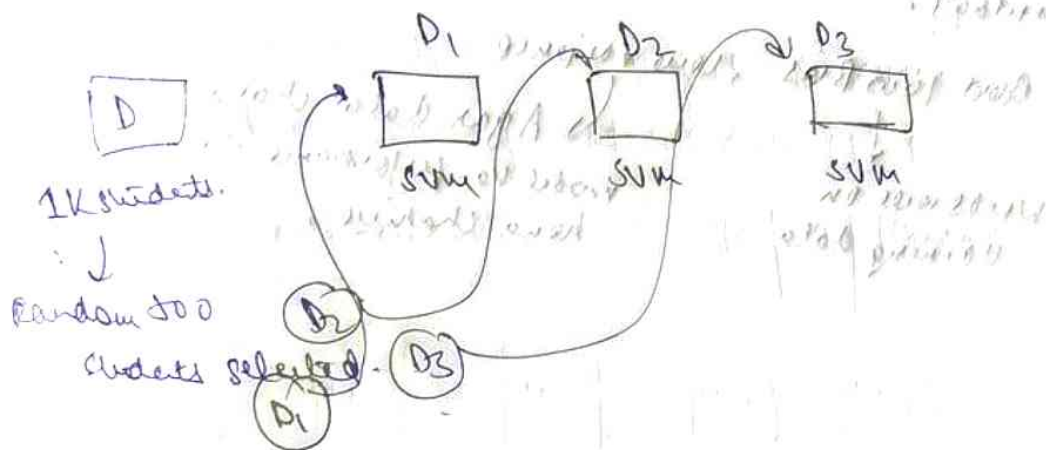
$\rightarrow$ VNN basis on this data and assigns weights

SUM	LOA	DT	Result
1	0	1	1
:	:	:	:

$(72) - 1 = 71$

* KNN decides weight of SVM, LR, DT

Bagging : (Bootstrapped aggregation) (Same model, diff data)



(Now among 12 slides, random 500 are selected $\rightarrow$ on

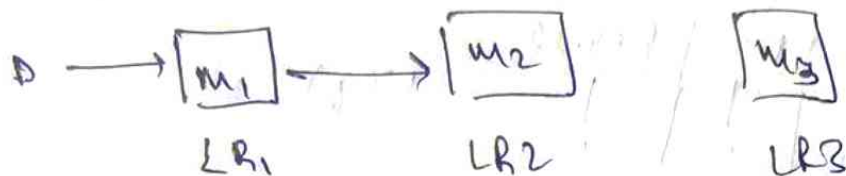
4. $\rightarrow D_3$

D_1 may or may not be D_1, D_2

X_9 is fit in m_1, m_2, m_3 and ϕ filter
majority ~~and~~ ϕ mean is calculated depending
on γ .

Random Forest: Base model is always Decision Tree

Boosting



- m1 works on data, notes the mistakes, pushes
- m2 works on same data, gives better results on mistakes, push to m3.

by the
better performances over

Advantages,

- ~~Good~~ low Bias, low variance

Works well on training data

→ Agar data change krle to model ka performance change krni karna chahie.